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Primary Topic	Theory: Aerosols
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How to Probe Cloud Microphysics from JWST Observations

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With the unprecedented accuracy of JWST, it is now possible to detect the presence of silicate clouds in exoplanet atmospheres by observing the spectral features of silicon-oxygen (Si-O) bonds in transmission and thermal emission spectra. Combining these observations with (micro-)physical cloud models allows us to investigate the cloud structures of exoplanets across a wide range of atmospheric environments. While the clouds of each planet are unique, they share the same underlying formation processes, allowing us to probe the physics of cloud formation.

We analyzed four JWST observations with confirmed cloud detections: WASP-107 b, WASP-17 b, VHS-1256 b, and YSES-1 c. By combining these observations with our cloud models, we uncovered a surprising similarity: the presence of small cloud particles above the main cloud deck. The origin of these particles is yet unclear. Two possible explanations arise: they may be the result of active aerosol formation, therefore offering insights into the first steps of cloud formation, or they may be advectively transported to higher altitudes, therefore providing clues about atmospheric circulation and climate dynamics. Thermal emission spectra offer an additional advantage: because clouds can determine the photosphere, these spectra are sensitive to the cloud particle accretion rate, which is regulated by the sticking coefficient (or contact angle). This allows us to probe the micro-physics of cloud formation in exoplanet atmospheres using panchromatic thermal emission spectra.

From just four exoplanets, we already find unexplained small particles in the upper atmosphere, shared cloud structures across hot Jupiters and wide-orbit companions, and new insights into the microphysics of cloud formation. With future JWST observations, we will be able to expand our analysis and investigate further cloud formation processes.

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